NATIONAL MATHEMATICS CONTEST PRIMARY-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SECTION A : 5 marks each

1. Arrange in ascending order: $\frac{5}{8}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{4}$
solution:
$\frac{1}{4}, \frac{1}{2}, \frac{5}{8}, \frac{2}{3}, \frac{3}{4}$

2. Eight apples weigh the same as two mangoes. Four mangoes weigh the same as two oranges. How many apples weigh the same as four oranges?

solution:
Let a, m and r represent apple, mangoes and oranges, respectively.
$8a = 2m$ and $4m = 2r$. It follows $8m = 4r$, Also $8m = 32a$. So $32a = 4r$. It is 32 apples.

3. How many four-digit numbers can you make using the digits $3, 5, 8$, and 4 if you can only use each digit once in each number?
solution:
$4! = 4 \times 3 \times 2 \times 1 = 24$ **Note:** Most candidates will list all the 24 four-digit numbers and that is okay.

4. Which of the following letters, A, B, C, D, E, H, I, M, N, O, P, T, U, V, W, X, Y, Z will look the same when placed in front of a mirror?
solution:
H, I, O, T, U, V, W, X, Y

5. Justine is twice as old as her brother Paul. In 5 years, Justine will be three times as old as Paul was 5 years ago. How old are Justine and Paul now?
solution:
Let the letter J and P represent Justine's and Paul's ages respectively. $J = 2P,\ J + 5 = 3(P - 5),\ P = 20, J = 40$

6. Sixteen children took part in a marathon. Annet was one of them. The number of children who finished after Annet was five more than the number who finished before her. In what position did Annet finish the marathon?

solution:
$x + 1 + x + 5 = 16$
After solving, $x = 5$
So, Annet was in the 6th position.

7. Andrew has a 7-litre and a 5-litre container but needs exactly 4 litres of water. He can only use these two containers and an unlimited supply of water. How can he measure exactly 4 litres?
solution:
   - *Fill the 7-litre container and pour the water from the 7-litre container into the 5-litre container until the 5-litre container is full which leaves 2 litres in the 7-litre container.*
   - *Empty the 5-litre container and pour the 2 litres from the 7-litre container into the empty 5-litre container.*
   - *Fill the 7-litre container again and carefully pour water from the 7-litre container into the 5-litre container (which already has 2 litres in it) until the 5-litre container is full.*
   - *Since the 5-litre container only had space for 3 more litres, there will be exactly 4 litres left in the 7-litre container.*

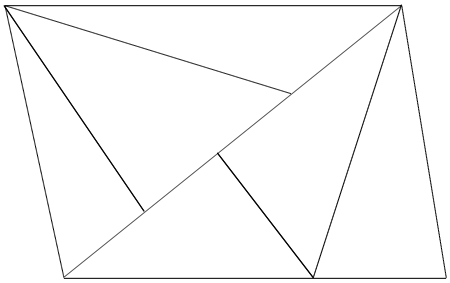
8. Two squares, each with a side length of 7 cm, overlap such that their overlapping region forms a square of 3 cm on each side. What is the total area covered by the two squares?
solution:
Area of one square $= 7\ cm \times 7\ cm = 49\ cm^2$. *Area of the overlapping square* $= 3\ cm \times 3\ cm = 9\ cm^2$. *The total area covered by both squares* $= 49\ cm^2 + 49\ cm^2 - 9\ cm^2 = 89\ cm^2$.

9. A farmer has ducks and rabbits. He counts 45 heads and 120 legs. How many ducks and how many rabbits does the farmer have?
solution:
Let d and r be the number of ducks and rabbits respectively. $d + r = 45,\ 2d + 4r = 120, r = 15, d = 30$

10. A shopkeeper mixes two types of rice. One type costs 5000 shillings per kg, and the other costs 3500 shillings per kg. If she wants to make a mixture that costs 4000 shillings per kg, in what ratio should she mix the two types of rice?
solution:
Let x be the amount (in kg) of the 5000 shillings/kg rice and let y be the amount (in kg) of the 3500 shillings/kg rice. The total cost of the mixture is the sum of the costs of each type of rice. The desired mixture cost is 4000 shillings/kg. So, $5000x + 3500y = 4000(x + y)$. *Solving yields* $1000x = 500y$. *Thus the ratio of* $x : y$ *is* $2 : 1$.

SECTION B : (10 marks each)

11. Count all the triangles in this figure.



solution:
There are 11 triangles and full marks should be given. Other candidates can score between 9 and 1 depending on the the number they have counted.

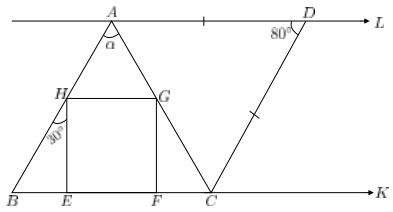
12. A frog starts at position 0 on a number line. It can only jump forward, to the right. On its first jump, it can jump 1 unit. On its second jump, it can jump 1 or 2 units. On its third jump, it can jump 1, 2, or 3 units, and so on. How many different sequences of jumps can the frog take to reach position 5?
solution:
The following are the possible jumps:

1) $1 \to 2 \to 3 \to 4 \to 5$
2) $1 \to 2 \to 3 \to 5$
3) $1 \to 2 \to 4 \to 5$
4) $1 \to 2 \to 5$
5) $1 \to 3 \to 4 \to 5$
6) $1 \to 3 \to 5$

They are **6** ways.

13. Two cyclists start from two towns 70 km apart and ride towards each other. One rides at 12 km/h and the other at 20 km/h. If they start 10 minutes apart, how long will it take before they meet?
solution:
They are riding towards each other. So the relative speed is $12\ km/h + 20\ km/h = 32\ km/h$. *The first cyclist starts 10 minutes earlier. In hours - 1/6 hours. In that time, the first cyclist travels:* $12km/h * (1/6)h = 2\ km$. *The remaining distance between them when the second cyclist starts is* $70\ km - 2\ km = 68\ km$. *The time it takes for them to meet* $= 68\ km/32\ km/h = 2.125$ *hours which is 2 hours and 7.5 minutes. It will take 2 hours and 7.5 minutes to meet after the second cyclist starts. For the first cyclist, it will take 2 hours and 17.5 minutes.*

14. In the figure on the right, triangle ADC is isosceles, $\angle BHE = 30°$, $\angle ADC = 80°$, line L is parallel to line K, rectangle $HGFE$ is inscribed inside $\triangle ABC$. Find the value of angle α.



solution:
Since $\triangle ADC$ *is isosceles with* $\angle CAD = \angle ACD$, $\angle ACD = \frac{180° - 80°}{2} = 50°$. *It follows that* $\angle CAD = 50°$
Since line L is parallel to line K, $\angle ACB = \angle CAD = 50°$.
Now $\angle ABC = \angle HBE = 90° - 30° = 60°$.
Therefore $\angle BAC = \alpha = 180° - (60° + 50°) = 70°$.

15. A special clock runs at a speed that is 30% faster than a normal clock. It was set correctly at midnight.

(a) What will the clock show when the actual time is 6:00 PM on the same day?

(b) If the clock was observed to show exactly 12:00 PM (noon) at some moment, what was the actual time at that instance?

solution:
(a)The special clock runs 30% faster, meaning it covers 1.3 hours in the same time a normal clock covers 1 hour. From midnight to 6:00 PM, 18 hours have passed. Time elapsed on the special clock: $18\ hours \times 1.3 = 23.4\ hours$. *Since the clock was set at midnight, it will show* 23.4 *hours past midnight. This is equivalent to* $11:24$ *PM. Therefore, when the actual time is 6:00 PM, the special clock will show 11:24 PM.*
(b) Time shown on the special clock: 12 *hours have passed since midnight. Equivalent time on a normal clock:* $12\ hours/1.3 \approx 9.23\ hours$. *9.23 hours past midnight is approximately 9:14 AM. Thus when the special clock shows 12:00 PM (noon), the actual time is approximately 9:14 AM.*

END